

(A) Dense update at $20\times$



all coordinates receive the step

Mean4 0.340 | severe degradation

(B) Dynamic Advantage Spine



per-tensor top 40% of adaptive Adam direction

support refreshed every optimization step

SCHEMATIC — dot positions are illustrative, not measured coordinates

(C) Observed at the tested limit

coordinates retained	39.3%
direction energy retained	85.0%
support Jaccard	0.600 ± 0.010
Dense $20\times$	0.340 $\rightarrow$ 0.445

Mean4 gain vs dense $20\times$: +0.105